

Exp:5 Greedy Best First Search

Program:

```
import heapq

def greedy_best_first_search(graph, heuristic, start, goal):
    priority_queue = [(heuristic[start], start)]
    visited = set()

    while priority_queue:
        h, node = heapq.heappop(priority_queue)

        if node in visited:
            continue

        print(node, end=" ")
        visited.add(node)

        if node == goal:
            print("\nGoal Reached!")
            return

        for neighbour in graph[node]:
            if neighbour not in visited:
                heapq.heappush(priority_queue, (heuristic[neighbour], neighbour))

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}

heuristic = {
    'A': 6,
```

```

        'B': 4,

        'C': 5,

        'D': 3,

        'E': 2,

        'F': 4,

        'G': 0

    }

    greedy_best_first_search(graph, heuristic, 'A', 'G')

```

Output:

```

Exp-5 Greedy Best First Search program.py - C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-5 Gr...
File Edit Format Run Options Window Help

# Greedy Best First Search Algorithm
def greedy_best_first_search(graph, heuristic, start, goal):
    priority_queue = [(heuristic[start], start)]
    visited = set()

    while priority_queue:
        h, node = heapq.heappop(priority_queue)

        if node in visited:
            continue

        print(node, end=" ")
        visited.add(node)

        if node == goal:
            print("\nGoal Reached!")
            return

        for neighbour in graph[node]:
            if neighbour not in visited:
                heapq.heappush(priority_queue, (heuristic[neighbour], neighbour))

# Graph Representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}

# Heuristic Values
heuristic = {
    'A': 6,
    'B': 4,
    'C': 5,
    'D': 3,
    'E': 2,
    'F': 4,
    'G': 0
}

greedy_best_first_search(graph, heuristic, 'A', 'G')
Ln: 48 Col: 0

```

```

IDLE Shell 3.13.5
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-5 Greedy Best First Se
arch/Exp-5 Greedy Best First Search program.py
A B E G
Goal Reached!
>>>
Ln: 7 Col: 0

```